SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Challa Ravi Teja 112325065 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ch Ravi Teja

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1Q Scenario: Event Communication Security

a. Encrypting the message using a shift of 4 positions

The Ceasar cipher is a classical encryption technique in which each letter of the plaintext is shifted by a fixed number of positions in the alphabet.

In this scenario, the shift value is $K = 4$

The encryption foomula is:

$$C = (P + K) \bmod 26$$

Here,
P represents the numerical possition of a plane text letter, C represents the ciphertent position, and K is the shift value

The Original Message is:
MEET ME AT NOON

Each letter is shifted four positions forward in the alphabet.

Plaintext	M	E	E	T	A	T	N	O	O	N
Shifted letter	Q	I	I	X	E	X	R	S	S	R

→ Spaces are preserved without encryption

Encrypted Message: QIIX EX RSSR

b. Decrypting the Message:

An authorized message member who knows that the messages uses a Ceasar cipher with a shift of 4 can recover the original message by shifting every encrypted later four positions backward.

The decryption formula is

$$P = (C - K) \bmod 26.$$

For example

* Q shifted backward by 4 positions become M.
* I shifted backward by 4 positions becomes E.
* X shifted backward by 4 positions become T.

Apply this rule to the entire ciphertext gives.

QIIX EX RSSR.

MEET AT NOON

The authorized member can therefore retrive the original event details.

Conclusion:

The ceasar cipher provides a simple method of the encoding messages. However, because it has only 25 possible non-zero shifts, it is not suitable for protecting highly confedential information against modern attackers.

2. Scenario: Intercepted Suspicious Message

a. Deduce the Most likely Original Message.

The intercepted message is

GSRH RH Z HVXIVG

The message appears to use the Atbash cipher, a classical substitution cipher in which the alphabet is reversed. Each letter is replaced by its corresponding letter from the opposite end of the alphabet

The substitution pattern is:

plaintext	Ciphertext
A	Z
B	Y
C	X
D	W
E	V
F	U
G	T
H	S
I	R
J	Q
K	P
L	O
M	N

→ The decrypt the intercepted message, apply the same reverse-alphabet mapping.

Ciphertext	plaintext
GSRH	THIS
RH	IS
Z	A
HVXING	SECRET

Therefore, The Original message is: THIS IS A SECRET

b. Justification using letter structures and Cipher character.

The Atbash cipher is a suitable explanation for the intercepted message for the following reasons.

1. Consistent substitution pattern.

Every letter is replaced according to a fixed reverse alphabet mapping. for example, G maps to T, S maps to H, and R maps to I.

2. Word - length preservation.

The ciphertext contains words of length 4, 2, 1 and 6. The decrypted message has excatly the same word lengths.

GSRH RH Z HVXING
THIS IS A SECRET

3. Repeating - letter patterns.

The word HVXING has no repond letters, and its depend SECRET, also has no repeated letters.

4. Meaningful plaintext.

The decrypted text forms a grammatically meaningful English sentence, which strongly supports the Atbash interpretation.

Q. Scenario: Secure Messaging System

a. Step by Step encryption of "HELLO" using the keyword "KEY".

The Company uses a poly alphabetic substitution cipher, commonly implemented using the Vigenere cipher.

This method uses a keyword to determine how much each plaintext letter is shifted.

The plaintext is HELLO, and the keyword is KEY

The Encryption formula is:

$$C = (P + K) \bmod 26.$$

Here:

* P is the Numerical value of a plaintext letter
* K is the Numerical value of the corresponding key letter
* C is the resulting cipher text value.

→ The letters are represented by numbers from 0 to 25 where, A=0, B=1, ---Z=25.

Step 1: Repeat the keyword

→ The keyword is shorter than the plaintext, so it is repeated until both have the same length.

plaintext	H	E	L	L	O
Repeated key.	K	E	Y	K	E

Step-2: Convert the letters into Keyvalues.

Character	H	E	L	L	O
plaintext value	7	4	11	11	14
Key character	K	E	Y	K	E
Key value	10	4	24	10	4

Step-3: Apply the encryption formula

Calculation	Result	Ciphertext
(7+10) mod 26	17	R
(4+4) mod 26	8	I
(11+24) mod 26	9	I
(11+10) mod 26	21	V
(14+4) mod 26	18	S

Therefore, The encrypted message is:

HELLO → RIIVS

b. Influence of the keyword repeating.

The repeating keyword determines the shift applied to each character of the plaintext. In this example, the key KEY is repeated as KEYKE.

Each letter of the key produces a different shift:

* K shifts the corresponding plaintext letter by 10 positions.
* E shifts by 4 positions.
* Y shifts by 24 positions.

Conclusion:

The vigenere ciphere Succesfully encrypts HELLO into RIIVS cuing the keyword KEY

Q. Scenario: Suspicious Digital Asset Investigation

a. Detecting hidden data within an image.

During a digital forensic investigation, the investigators suspect that a company logo contains hidden confidential information. This process may involve steganography which is the technique of concealing information within an ordinary-looking file, such as image, audio recording, or video.

1. Visual inspection:

Investigators examine the image for unusual colors, distorted regions, unexpected patterns, or suspicious pixel differences. However, sophisticated steganography may not produce visible changes.

2. Metadata analysis.

The images metadata can be inspected for unusual comments, unexpected fields, editing history, or information that does not match the file's claimed origin.

3. File structure analysis.

Investigators examine the image file structure to identify unexpected data appended after the normal image content, unusual chunks.

4. Statistical analysis:

Steganography tools may alter pixel value or colour distributions. Investigators can use statistical tests to identify unusual patterns.

5. Comparision with the Original image.

If an authentic copy of the company logo is availble investigators can compare the two images pixel by pixel Unexpected differences may indicate notification or the embedding of additional information.

6. Steganography detection tools.

Specilized forensic tools can analyze image characterstics, metadata, and pixel distributions to identify possible hidden content.

b. Two techniques to strengthen the Security of hidden Communications.

1. Encrypt the confidential data before embedding it.

The confidential message should first be encrypted using a strong, modern encryption algorithm such as AES before being embedded into the image.

2. Use robust Steganography embedding with integrity protection.

The Sender can use a well designed steganographic method that distributes the hidden data across Suitable image regions while minimizing detectable statical change.

Conclusion:

Digital forensic investigators can detect potential image-based Steganography through visual, structural, metadata, and statistical analysis.

Q2 Scenario: Legacy Encryption in corporate Communication

A. Recovering the Original message

The received Ciphertext is:

WECRLTEERDSOEEFEAOCAIVDEN

The Scenario describes a transposition-based encryption method with three levels of arrangement. The Ciphertext is consistent with the three-rail Rail Fence cipher, a transposition cipher in which letter are written diagonally across three rails and then read row by row.

Step-1: Understand the encryption method.

In a three-rail Rail Fence cipher, plaintext character are arranged in a zigzag pattern across three rows. For example, The begining of the plaintext is arranged as follows:

Three-rail zigzag arrangement

W E C
 E R L

Step 2: Reverse the Transposition.

To decrypt the message, the receiver determines the zigzag positions for the ciphertext length, divides the ciphertext among the three rails.

Step 3: Read the plaintext

Original Message: WEAREDISCOVEREDFLEEATONCE

"WE ARE DISCOVERED FLEE AT ONCE"

b. Why is this method vulnerable in modern Communicate Systems ?

The three-rail Rail Fence cipher is unsuitable for modern corporate communications for several reasons.

1. It does not change the letters.

The cipher only rearranges the positions of the original letters. All plaintext characters remains present in the ciphertext, which limits its security.

2. Small key Space

A Three-rail cipher uses a very simple arrangement. An attackers can test a small number of possible rail counts and identify a meaningful plaintext relatively quickly.

3. Vulnerability to known plaintext attacks.

If an attacker knowns or guesses part of the original message, They can compare it with the ciphertext to infer the transposition pattern.

4. unsuitable for Senitive corporate information.

Confedential business plains, financial information, employee records, and intial credantials require stronger protection than a simple transposition cipher can provide.

Conclusion:

The ciphertext can be decrypted using the three-rail Rail Fence cipher to recover WE ARE DISCOVERED FLEE AT ONCE.